



Stability and Routh Criterion

As we know a system is stable if the roots of the denominator polynomial has negative real parts.

BIBO Stability: Bounded input bounded output stability:

The impulse response $h(t)$ is BIBO stable if and only if

$$\int_{-\infty}^{\infty} |h(t)| dt < \infty \quad \text{or finite.}$$

Routh criteria:

$$a(s) = s^n + a_1 s^{n-1} + a_2 s^{n-2} + \dots + a_n$$

$a(s)$ is a polynomial of order n .

The denominator of the closed loop transfer function can be represented by

$a(s)$.

Routh said that a necessary and sufficient condition for stability is that the first column of Routh array is positive. Routh table is formed in the following way:

$$\begin{array}{lcll} s^n & : & 1 & a_2 \quad a_4 \quad \dots \\ s^{n-1} & : & a_1 & a_3 \quad a_5 \quad \dots \\ s^{n-2} & : & b_1 & b_2 \quad b_3 \quad \dots \end{array}$$

$$\begin{array}{c}
 s^{n-3} : c_1 \quad c_2 \quad c_3 \quad \dots \\
 \vdots \\
 s^1 : * \quad * \\
 s^0 : *
 \end{array}
 \rightarrow \text{positive}$$

$$b_1 = \frac{a_1 a_2 - a_3}{a_1}, \quad b_2 = \frac{a_1 a_4 - a_5}{a_1}$$

$$b_3 = \frac{a_1 a_6 - a_7}{a_1}, \quad c_1 = \frac{b_1 a_3 - a_1 b_2}{b_1}$$

$$c_2 = \frac{b_1 a_5 - b_3 a_1}{b_1}, \quad c_3 = \frac{b_1 a_7 - b_4 a_1}{b_1}$$

Let's see an example:

$$a(s) = s^6 + 4s^5 + 3s^4 + 2s^3 + s^2 + 4s + 4$$

we need to determine if any of the roots are in the right half plane without solving it.

Routh table:

s^6	:	1	3	1	4
s^5	:	4	2	4	0
s^4	:	$\frac{5}{2}$	0	4	
s^3	:	2	$-\frac{12}{5}$	0	
s^2	:	3	4		
s^1	:	$-\frac{76}{15}$	0		
s^0	:	4			

calculations:-

$$\frac{5}{2} = \frac{4 \cdot 3 - 1 \cdot 2}{4}, \quad 0 = \frac{4 \cdot 1 - 4 \cdot 1}{4}$$

$$4 = \frac{4 \cdot 4 - 1 \cdot 0}{4}, \quad 2 = \frac{\frac{5}{2} \cdot 2 - 4 \cdot 0}{\frac{5}{2}}$$

$$-\frac{12}{5} = \frac{\frac{5}{2} \cdot 4 - 4 \cdot 4}{\frac{5}{2}}$$

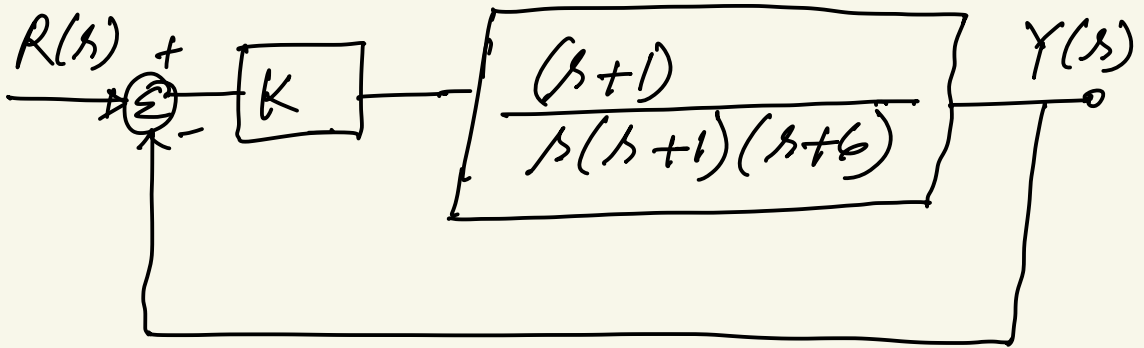
$$3 = \frac{2 \cdot 0 - \left(\frac{5}{2}\right)\left(-\frac{12}{5}\right)}{2}$$

$$4 = \frac{2 \cdot 4 - \left(\frac{5}{2} \cdot 0\right)}{2}$$

$$-\frac{76}{15} = \frac{3\left(-\frac{12}{5}\right) - 8}{3}$$

$$4 = \frac{\left(-\frac{76}{15}\right) \cdot 4 - 0}{\left(-\frac{76}{15}\right)}$$

Consider the following closed-loop feedback system



$$T_c(s) = \frac{k \cdot \frac{(s+1)}{s(s+1)(s+6)}}{1 + k \frac{(s+1)}{s(s+1)(s+6)}}$$

after simplification:

$$T_c(s) = \frac{k(s+1)}{s^3 + 5s^2 + (k-6)s + k}$$

Denominator of the closed-loop transfer function :

$$a(s) = s^3 + 5s^2 + (k-6)s + k$$

Form Routh table :

$s^3 :$	1	$k-6$	0
$s^2 :$	5	k	0
$s^1 :$	$\frac{4k-30}{5}$	0	0
$s^0 :$	k	0	0

In order for the system to be stable

$$\frac{4k-30}{5} > 0, \quad k > 0$$

$$\Rightarrow 4k - 30 > 0$$

$$\Rightarrow k > \frac{15}{2}$$

Matlab code demonstrated in
the class.